Tea & Toast

$E_w = F\ d$	$F = m\ g$	$g = 9.8N/kg$



Toast

1. State the energy forms for the input and output of the kettle.
2. Draw a universal system diagram for toaster.
3. Calculate the work done by a person to carry the toaster from the D&T department upstairs to the Science department.

Tea

1. Draw a sub-system system diagram for kettle. Include any relevant feedback loop.
2. Describe, by referencing your sub-system diagram, the operation of the kettle. Remember: SET – ACTION – FEEDBACK – COMPARE – ADJUST.
3. It requires 58.7J of energy to carry the empty kettle up the Art department, 7.68m above the D&T department.
   Calculate the mass of the empty kettle.
4. Describe how the kettle design could be changed to make it more energy efficient.